

**TripTrek: Intelligent Travel Planning Using PaLM's Chat-
Baison-001**

An Industrial Internship Report

Submitted by

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ABSTRACT

Travel planning is a structured process aimed at creating a smooth, enjoyable, and budget-friendly travel experience. It involves selecting a destination, setting a budget, organizing transportation, arranging accommodations, and creating a well-balanced itinerary of activities and relaxation. By researching factors like weather, local costs, and travel restrictions, travelers can make informed choices that align with their interests and resources. Thoughtful travel planning not only helps to optimize time and manage expenses but also minimizes unexpected challenges, allowing for a more immersive and rewarding journey. This project presents TripTrek, a conversational AI-powered travel planning system built upon PaLM's Chat-Bison-001 language model. TripTrek leverages the model's ability to process and understand natural language queries to provide personalized travel recommendations, itinerary generation, and real-time updates. The system integrates with various travel APIs and databases to offer comprehensive information on destinations, accommodations, flights, and activities. By combining the power of large language models with travel data, TripTrek aims to revolutionize the travel planning experience, making it more efficient, personalized, and enjoyable.

Keywords: Travel planning, conversational AI, large language models, PaLM's Chat-Bison-001, personalized recommendations, itinerary generation.

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1. Introduction:

1.1. Problem Statement:

TripTrek is an AI-powered travel planning platform designed to revolutionize the way people plan and organize their trips. By leveraging advanced artificial intelligence algorithms, TripTrek offers users personalized travel itineraries tailored to their preferences, interests, and budget constraints.

The various problems tackled by trip trek includes, A family of four is trying to plan an exotic vacation. They are facing difficulties in finding the right destinations, creating a suitable itinerary, and gathering sufficient information about the places they plan to visit. A manager on a business trip is looking for a suitable meeting venue to impress a client.

They are unfamiliar with the area and lack knowledge about potential meeting spots and daily planning. A teacher organizing an educational trip for students is struggling to find historical and culturally significant places. They have limited time and need to create a well-structured itinerary that covers multiple locations while ensuring the students learn about each place. All of them are provided with Travel itineraries to perfectly plan their trip through TripTrek.

1.2. Motivation:

The motivation behind TripTrek is to address the complexity and effort involved in planning travel experiences, making it easier, faster, and more enjoyable for users. The travel planning process often requires hours of research, filtering through vast amounts of information on accommodations, dining, activities, and transportation options to create a balanced itinerary that aligns with personal preferences, interests, and budget. For many, this process can become overwhelming, causing stress and diminishing the excitement of travel.

TripTrek seeks to solve these issues by leveraging advanced AI algorithms and machine learning to offer users a fully personalized, efficient, and enjoyable travel planning experience. With a combination of rich travel data and user-centered design, TripTrek aims to:

Streamline Trip Planning: By automating itinerary creation based on user preferences and available data, TripTrek significantly reduces the time and effort needed to plan a trip, allowing users to focus on enjoying the journey rather than planning.

Provide Personalization: Through AI-driven recommendations, TripTrek delivers tailored travel suggestions that align with individual interests, preferred activities, and budget, enhancing the overall relevance and satisfaction of the travel experience.

Enhance Travel Accessibility: By simplifying complex planning processes, TripTrek opens up travel opportunities for people who may have felt discouraged by the logistics of planning, thus making travel more accessible.

Offer a Comprehensive Experience: TripTrek consolidates information on accommodations, activities, dining, and transport, presenting users with a cohesive, easy-to-follow plan that covers all aspects of their trip.

Reduce Planning Stress: TripTrek empowers travelers to enjoy a seamless and stress-free experience by taking care of the details and providing peace of mind that their itinerary is well-organized and tailored to their needs.

In essence, TripTrek's motivation lies in transforming travel planning from a challenging task into an intuitive and enjoyable experience, using AI to deliver a higher quality of personalized service than traditional methods. This makes TripTrek not only a tool for organizing travel but a partner in curating unforgettable journeys.

1.3. Objective:

- Assist users in planning and booking their trips
- Provide personalized travel recommendations
- Offer real-time travel information and updates
- Help users make informed decisions about their travel plans Using Chat-Baison-001, By leveraging advanced artificial intelligence algorithms.

1.3.1. Proposed system: TRIP TREK

TripTrek offers users personalized travel itineraries tailored to their preferences, interests, and budget constraints by leveraging advanced artificial intelligence algorithms.

The platform combines machine learning models with rich travel data to provide users with comprehensive recommendations for accommodations, activities, dining options, transportation, and more. It suggests family-friendly attractions like amusement parks, museums, and scenic spots, and provides recommendations for nearby restaurants and cafes that cater to diverse dietary needs.

It recommends key business venues such as conference centers and meeting locations, along with local attractions for downtime. Additionally, it provides suggestions for nearby restaurants and cafes suitable for business lunches and dinners.

1.3.2. Advantages of proposed system:

- ✓ Tailored Recommendations
- ✓ Customizable Itineraries
- ✓ Time-Saving
- ✓ Easy-to-Use Interface
- ✓ Access to Real-Time Data
- ✓ Budget-Friendly Planning
- ✓ Reduced Stress
- ✓ Hassle-Free Travel

2. Technologies learnt:

2.1. Vertex AI:

Google Cloud's comprehensive machine learning (ML) and artificial intelligence (AI) platform, is particularly useful for building, deploying, and managing generative AI models. It provides an integrated environment for AI practitioners, allowing them to leverage Google's infrastructure, pre-trained models, and

advanced tools to build generative AI solutions with ease. Here's how Vertex AI facilitates learning and development with generative AI:

- **Access to Pre-trained Generative Models**

Vertex AI Model Garden: Vertex AI offers access to pre-trained generative models like PaLM, and code generation tasks. These models can serve as starting points for building applications such as chatbots, image generation, and code assistance.

- **Unified Workspace for Experimentation**

Vertex AI Workbench: Vertex AI Workbench is a collaborative and flexible Jupyter-based environment where you can design, test, and iterate on generative models. It allows you to experiment with generative AI workflows using notebooks that support various programming languages and libraries.

Data Management: Vertex AI provides integration with Google Cloud's data storage solutions, making it easier to manage, preprocess, and feed data into models.

- **Custom Model Training and Development**

Compute Resources: Vertex AI offers access to GPUs and TPUs, which are essential for efficiently training generative models with large datasets.

- **Prompt Engineering and Tuning**

Prompt Design Tools: Generative models rely on prompt engineering, especially when working with language models. Vertex AI provides tools and guidance for optimizing prompts, making it easier to get accurate and contextually relevant outputs from language models.

- **Pipeline Automation with Vertex Pipelines**

Vertex Pipelines: Automates the entire ML workflow, from data ingestion and preprocessing to model training, evaluation, and deployment. For generative AI,

this allows you to automate repetitive tasks, monitor performance, and iterate on the model quickly.

- **Evaluation and Interpretability Tools**

Explainable AI: Vertex AI provides explainability features that help interpret and explain model outputs, which can be particularly valuable for understanding generative models' behavior and identifying areas for improvement.

Evaluation Metrics: Vertex AI offers various metrics and evaluation tools for generative tasks, enabling users to assess model performance and make iterative improvements.

- **Collaboration and Learning Resources**

AI Learning Resources: Google Cloud offers a range of tutorials, labs, and courses focused on generative AI through Vertex AI. These resources provide a solid foundation in generative model concepts, practical applications, and Vertex AI tools, helping teams learn and upskill effectively.

Collaborative Development: Vertex AI integrates with Google Workspace, allowing teams to collaborate and share insights from experiments and development, making it a strong platform for learning generative AI as a team.

2.2. Visual Studio code:

VS Code (Visual Studio Code) is a powerful and popular code editor that is widely used for a variety of programming tasks, including developing and learning generative AI using platforms like Vertex AI. When it comes to learning and working with generative AI (such as PaLM or other language models) using Vertex AI, VS Code can be a central tool for interacting with code, managing projects, and leveraging AI/ML workflows.

- **Writing and Executing Python Code for AI Models**

AI Frameworks: Python is the most common programming language used for machine learning, and Vertex AI supports Python for model training, evaluation,

and deployment. VS Code allows you to write Python code for: Building and fine-tuning models. Working with libraries like TensorFlow, PyTorch, and scikit-learn (often used for training generative AI models).Implementing PaLM or other generative AI models using Vertex AI's APIs.

Jupyter Notebooks: VS Code supports Jupyter Notebooks via extensions, which is useful for exploratory data analysis (EDA), model training, and experimentation. Notebooks provide an interactive environment where users can experiment with code in a cell-by-cell manner, making it easier to debug and visualize outputs.

- Integrating Vertex AI API and SDKs

Vertex AI SDK: The Vertex AI SDK is a set of Python libraries provided by Google Cloud to interact with Vertex AI services (e.g., for model training, deployment, and monitoring). In VS Code, you can write code to:Authenticate with Google Cloud services (e.g., using google-auth), Train and fine-tune models using Vertex AI's managed infrastructure, Deploy models to Vertex AI Endpoints for serving predictions.

Command-Line Interface (CLI): VS Code's integrated terminal can be used to interact with Google Cloud SDK (gcloud CLI), which allows you to interact with Vertex AI resources, such as creating and managing model training jobs, data pipelines, and deployments, Working with Vertex AI Models.

Model Training: In VS Code, you can write scripts that interface with Vertex AI to submit training jobs. You can define custom training pipelines, handle datasets, and configure hyperparameters, all within your VS Code environment. You can also monitor the status of these jobs and retrieve model artifacts once training is complete.

Fine-tuning Pre-trained Models: For generative AI models like PaLM, you can fine-tune pre-trained models by writing training scripts directly in VS Code, which can then be executed using Vertex AI's managed training services. You can modify these models to fit your own dataset and task, such as adapting PaLM to a specific style of text generation.

Model Inference: Once your model is trained and deployed to Vertex AI Endpoints, you can write code in VS Code to perform inference on new data, generating predictions or content. This could include generating text based on input prompts or summarizing documents.

2.3. Streamlit

Using Streamlit for learning generative AI with Vertex AI focuses on building interactive web applications to showcase AI models.

Building Interactive Interfaces: Streamlit allows you to create user-friendly web apps where users can input text and get real-time AI-generated content, such as text generation from a model, like PaLM deployed via Vertex AI.

Model Integration: You learn how to connect Streamlit apps to models hosted on Vertex AI by calling APIs for inference and displaying results, such as text completions or summaries, directly in the app.

Customizing AI Models: You can add features to let users adjust model parameters (e.g., creativity, text length) and experiment with different outputs.

Real-Time Feedback: Streamlit provides automatic updates, ensuring a smooth user experience by displaying AI-generated content instantly.

Deploying Models: You learn how to deploy your Streamlit app and connect it to cloud infrastructure, making the generative AI model accessible to users worldwide.

Visualization: Streamlit supports visualizing model outputs (text, metrics) and integrates with libraries like Matplotlib for data visualization.

2.4. Google cloud

Use Google Cloud Storage to securely store large training datasets and model outputs. Big Query enables fast data querying and pre-processing to prepare data for model training. Together, they simplify data access, storage management, and preprocessing in the cloud.

Model Training

Vertex AI provides tools for training, fine-tuning, and evaluating generative models. Supports Auto ML and custom model training with optimized infrastructure. Easily scale up resources (e.g., GPUs/TPUs) to handle complex model requirements.

Deployment

Deploy models as APIs with Vertex AI Endpoints for scalable, real-time predictions. Kubernetes Engine can be used for complex or containerized model deployment setups. Both options allow seamless model access for web apps or other user-facing services.

Compute Resources

Compute Engine provides customizable VMs for specific training or deployment needs. Scale workloads using GPUs/TPUs to meet high-performance requirements. Kubernetes Engine supports containerized deployments for efficient model scaling.

Pre-trained APIs

Google's APIs, like Natural Language and Vision API, complement generative models. These APIs offer ready-to-use functions for NLP, image processing, and more.

2.5. PaLM API

a. Understanding Large Language Models (LLMs)

Learn about the transformer architecture that drives language models like PaLM, focusing on the self-attention mechanism. Understand how LLMs process text data and predict sequences in a context-aware manner. Explore why transformers are effective for natural language tasks, enabling text generation, summarization, and question-answering.

b. Prompt Engineering for Generative AI

Experiment with crafting prompts to guide model behavior and generate coherent, context-relevant responses. Learn about prompt tuning to improve outputs across different tasks, such as text completion and summarization. Recognize the impact of prompt structure on the quality and relevance of generated content.

3. System Design:

3.1. System Architecture:

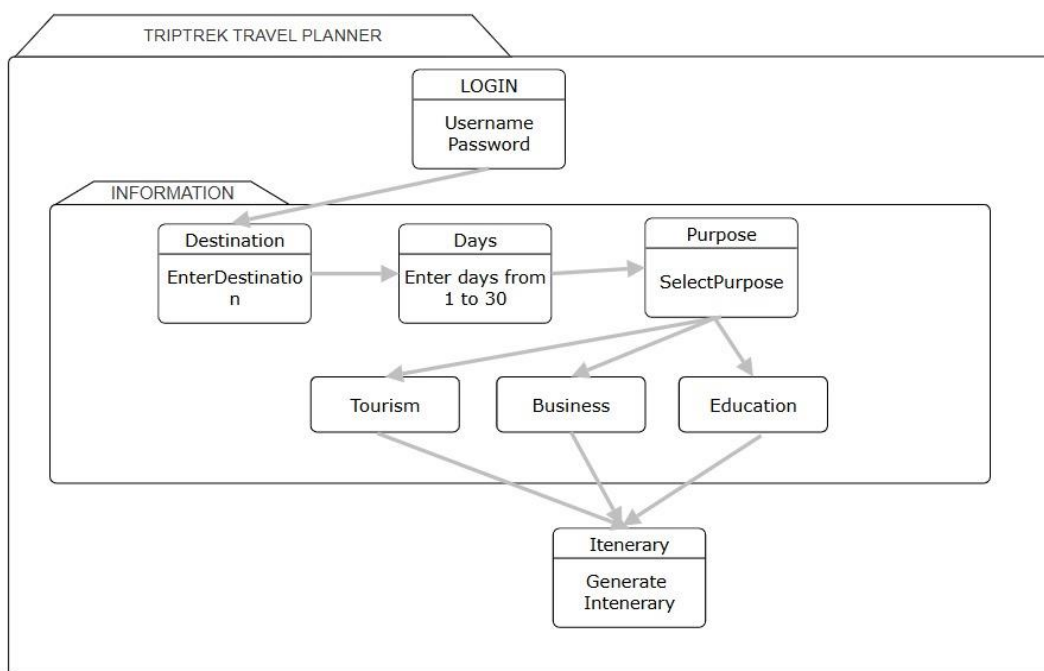


Fig 3.1 TripTrek System Architecture

The diagram illustrates the architecture of our travel planning system, TripTrek. Here's a breakdown of its components and how they interact:

a. **Login:** Users start by logging into the system with their username and password.

b. **Information:** Once logged in, users provide essential information for their trip:

- **Destination:** They enter the destination they want to visit.
- **Days:** They specify the number of days they plan to stay (between 1 and 30).

- **Purpose:** They select the purpose of their trip from three options: Tourism, Business, or Education.

b. **Itinerary Generation:** Based on the information provided, the system generates a customized itinerary. This is the core functionality of TripTrek.

b. **Purpose-Specific Recommendations:** The system can also provide recommendations tailored to the selected purpose:

- **Tourism:** Suggestions for tourist attractions, activities, and local experiences.
- **Business:** Recommendations for meeting venues, business centers, and networking opportunities.
- **Education:** Suggestions for educational institutions, historical sites, and cultural landmarks.

Overall, the TripTrek system aims to simplify travel planning by providing personalized itineraries and recommendations based on user preferences and the purpose of their trip.

3.2. Target performance metrics

- Number of new users per month
- User acquisition cost (CAC)
- Monthly active users (MAU)
- Customer satisfaction score (CSAT)
- User feedback and reviews

3.3. Gaps Addressed by the Proposed System

TripTrek aims to address several gaps in the existing travel planning landscape:

a. Personalization:

Tailored Recommendations: Traditional travel planning tools often provide generic recommendations. TripTrek leverages AI to offer personalized suggestions based on individual preferences, interests, and past travel history.

b. Efficiency:

Automated Planning: TripTrek automates many time-consuming tasks, such as researching destinations, booking flights and accommodations, and creating itineraries.

Real-Time Updates: The system can access real-time data on flights, accommodations, and attractions, ensuring that users have the most up-to-date information.

c. Comprehensive Information:

Expert Insights: TripTrek can provide expert insights and advice on travel destinations, cultural nuances, and safety tips.

Local Knowledge: The system can leverage local knowledge and recommendations from experienced travelers.

d. Cost-Effectiveness:

Budget-Friendly Planning: TripTrek can help users find the best deals on flights, accommodations, and activities, saving them money.

Reduced Stress: By streamlining the planning process, TripTrek can help reduce stress and anxiety associated with travel.

3.4. Module Description:

This is the details of the evaluation of models for TripTrek, a platform leveraging AI, designed to revolutionize the way people plan and organize their trips

a. Template Based Approach:

- Employs pre-defined templates of itineraries, with placeholders for user information. Human evaluation score of generated materials.
- Hyperparameters
 - Set of pre-defined templates for different job types.
 - Weighting of keywords in user profiles for template selection.
- Performance: Accuracy score = 81%

b. Keyword Focused Approach

- Generates itineraries by inserting user-interest keywords into a base template. Human evaluation score of generated materials.
- Hyperparameters:
 - Keyword extraction algorithms for user purpose and destination.
 - Scoring function for keyword relevance.\
- Performance: Accuracy score = 73%

c. Transformer -based Approach (e.g., GPT3, PaLM)

- Utilizes a pre-trained large language model (LLM) to generate personalized itineraries based on user input and destination descriptions. Human evaluation score of generated materials.
- Hyperparameters:
 - Fine-tuning the LLM on travel data.
 - Hyperparameters specific to the chosen LLM
- Performance: Accuracy score = 81%

Thus, from these models the transformer-based approach is followed in this project to utilize the PaLM's chat-baison-001 model to generate personalized content for travel itinerary

3.5 System Specification:

To develop a solution that addresses the problem of complex travel itinerary planning, certain software and hardware specifications are required. These specifications ensure that the system operates efficiently, is accessible to users, and provides a seamless experience.

3.5.1 Software Requirements:

- Frameworks: Python frameworks, Streamlit
- Libraries: Streamlit, generative-ai Altair
- Development Environment: VSCode

3.5.2 Hardware Requirements:

- Computing resources: T4 GPU CPU/GPU specifications, number:
- Memory: RAM 8 GB
- Storage: 1 TB SSD Disk space for models and logs

3.6 Detailed Design:

3.6.1 Use case Diagram:



Fig 3.6.1 TripTrek Use case Diagram

The provided use case diagram depicts a travel booking system with multiple user roles and their interactions with the system. Users can sign up, log in, view and edit their itineraries, get recommendations, and provide trip preferences. They can also book accommodations and transportation, review and rate their experiences, and provide feedback.

3.6.2 Sequence Diagram:

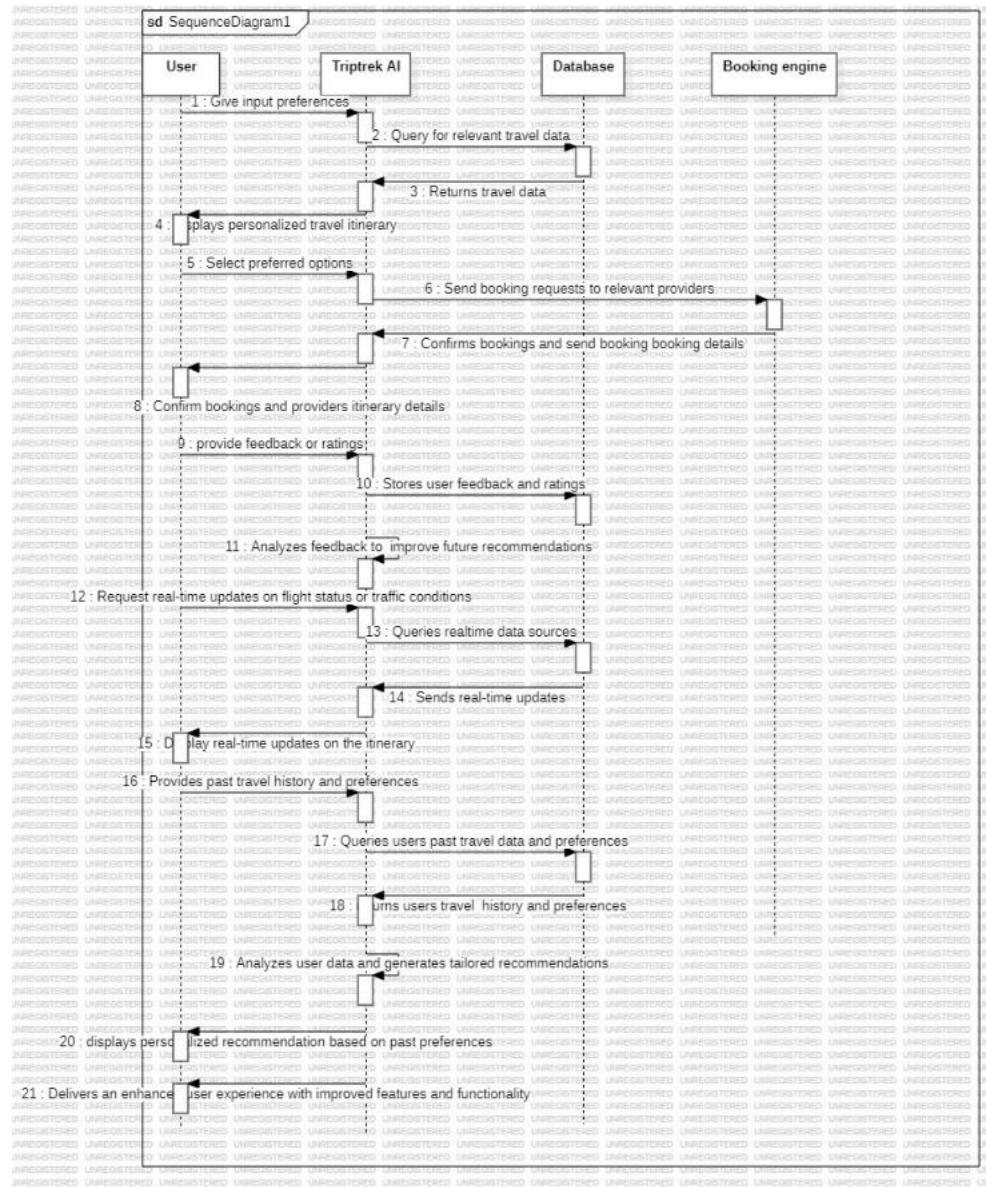


Fig 3.6.2 TripTrek Sequence Diagram

3.6.3 Class Diagram:

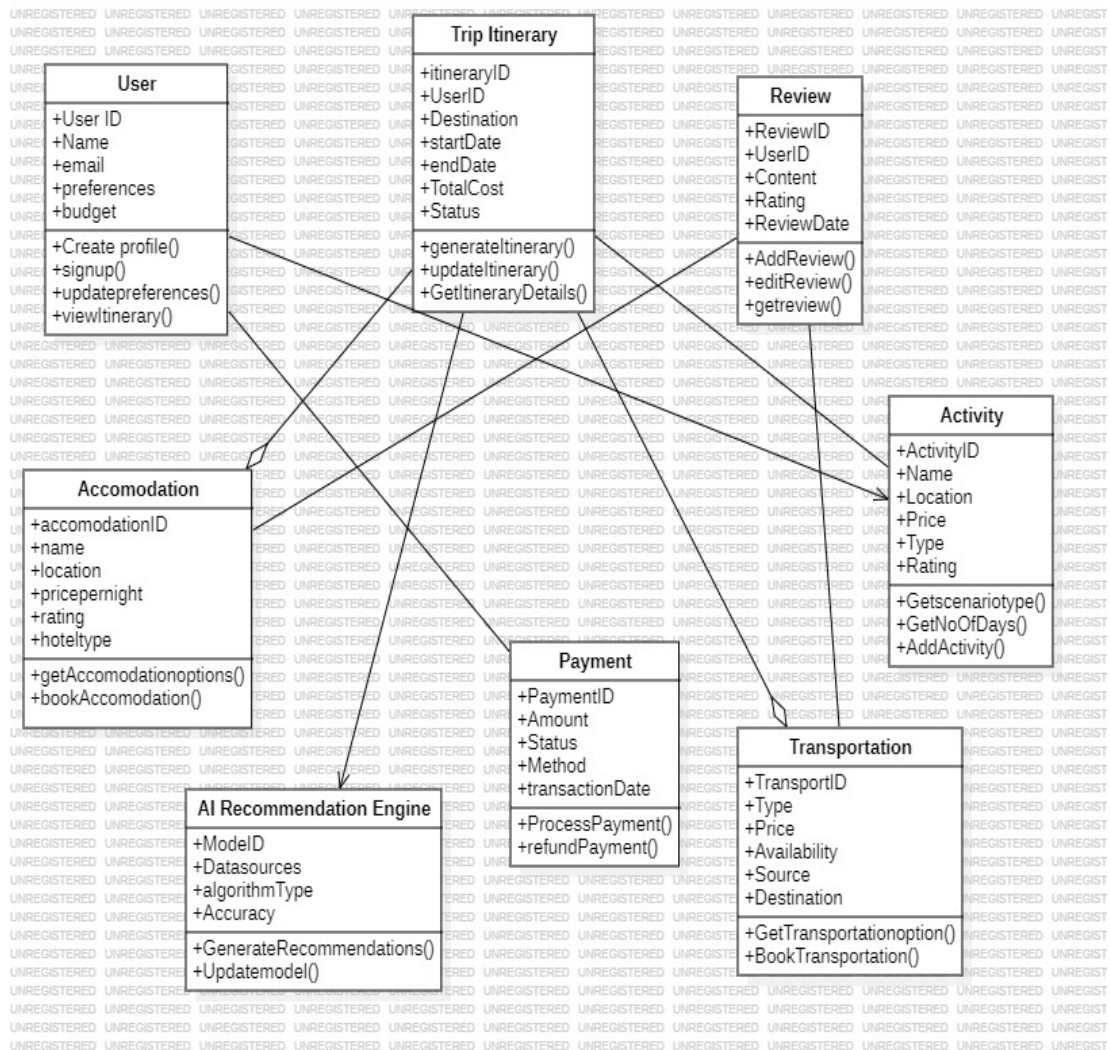


Fig 3.6.3 TripTrek Class Diagram

The system establishes various relationships between entities. Users can create multiple trip itineraries, provide reviews, and make payments. Trip itineraries are associated with users, accommodations, transportation options, and payments. Reviews are linked to specific users and their corresponding trip itineraries. Accommodations are connected to trip itineraries and have associated ratings. Payments are linked to both trip itineraries and users. Transportation options are connected to trip itineraries and indicate their availability. The AI recommendation engine utilizes user preferences and trip data to generate tailored recommendations.

3.6.4 Dataflow Diagram:

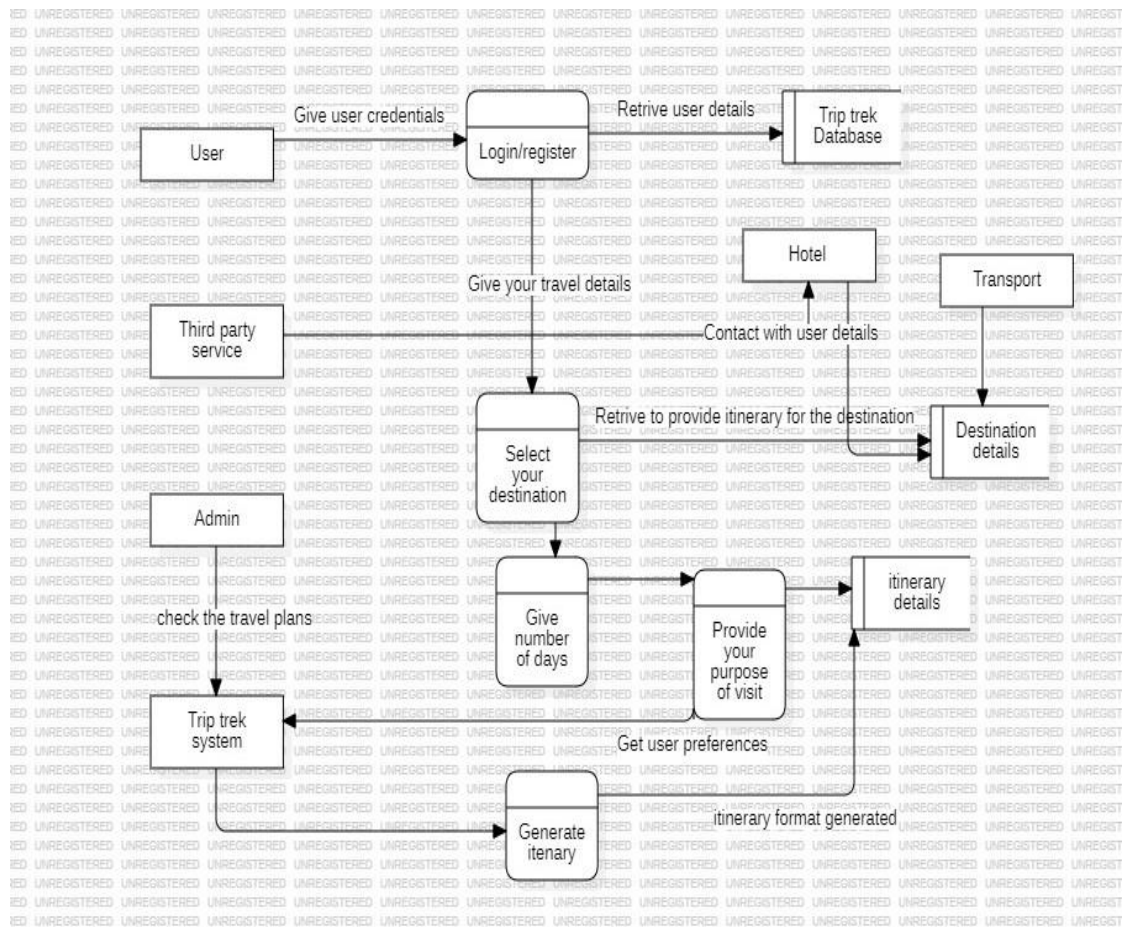


Fig 3.6.4 TripTrek Data Flow Diagram

The dataflow diagram illustrates a travel planning system called Trip Trek. It outlines the flow of data and information throughout the system, involving various actors and processes. Users begin by providing their credentials to login or register. Once authenticated, they enter their travel details, including destination, number of days, and purpose of visit. The system retrieves user preferences and generates an itinerary based on the provided information. The generated itinerary is then presented to the user in a suitable format.

3.6.5 Activity Diagram:

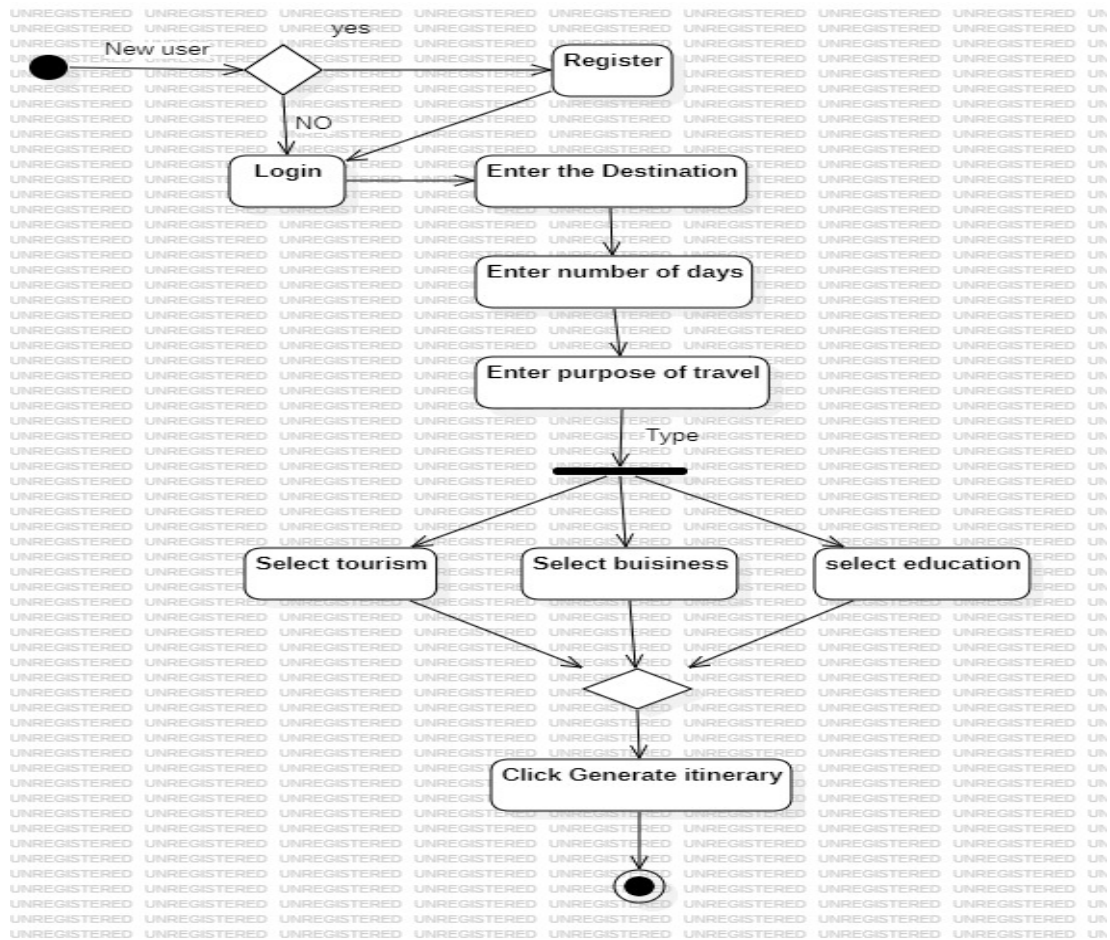


Fig 3.6.5 TripTrek Activity Diagram

The activity diagram illustrates the process of generating a travel itinerary within a system. It starts with a new user who either registers for an account or logs in if they already have one. Once logged in, the user is prompted to enter their desired destination, the number of days for the trip, and the purpose of their visit (tourism, business, or education). Based on the selected purpose, the system guides the user through specific details relevant to their chosen category. After providing all necessary information, the user clicks on the "Generate Itinerary" button. The system then processes the input data and generates a customized itinerary tailored to the user's preferences and requirements. The process concludes with the system presenting the generated itinerary to the user.

4 Implementation

4.1. Implementation Details:

In VSCode Create the Project folder, TRAVELPLANNER which contains files as shown below:

1. app.py: It serves as the primary application file housing both the model and Streamlit UI code.
2. requirements.txt: It enumerates the libraries necessary for installation to ensure proper functioning.

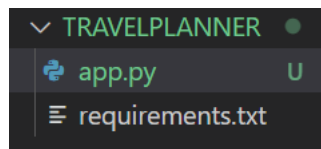


Fig 4.1.1 TripTrek Project Folder

Milestone 1: Requirements specification

Specifying the required libraries in the requirements.txt file ensures seamless setup and reproducibility of the project environment, making it easier for others to replicate the development environment.

Activity 1: Create a requirements text file to list the required libraries

Streamlit: Streamlit is a powerful framework for building interactive web applications with Python.

google-generativeai: Python client library for accessing the GenerativeAI API, facilitating interactions with pre-trained language models like chat-bison-001 .

altair<5 : Python library for data visualization, providing a simple and intuitive API for creating interactive visualizations, especially for datasets smaller than 5 million rows (Altair<5).

```
TRAVELPLANNER > requirements.txt
1
2 #libraries to be installed
3 streamlit==1.10.0
4 google-generativeai
5 altair<5
6 streamlit-option-menu
```

Fig 4.1.2 TripTrek Requirements Folder

Activity 2 : Install the required libraries

Open the terminal. Run the command: `pip install -r requirements.txt` This command installs all the libraries listed in the requirements.txt file

```
PS C:\Users\senil\Hello-world\TRAVELPLANNER> pip install -r requirements.txt
Requirement already satisfied: streamlit==1.10.0 in c:\users\senil\anaconda3\envs\myenv\lib\site-packages (from -r requirements.txt (line 3)) (1.10.0)
Requirement already satisfied: google-generativeai in c:\users\senil\anaconda3\envs\myenv\lib\site-packages (from -r requirements.txt (line 4)) (0.7.2)
Requirement already satisfied: altair<5 in c:\users\senil\anaconda3\envs\myenv\lib\site-packages (from -r requirements.txt (line 5)) (4.2.2)
```

Fig 4.1.3 TripTrek Libraries Installation

Milestone 2: Initializing the PALM The Google API key

PALM The Google API key is a secure access token provided by Google, enabling developers to authenticate and interact with various Google APIs. It acts as a form of identification, allowing users to access specific Google services and resources. This key plays a crucial role in authorizing and securing API requests, ensuring that only authorized users can access and utilize Google's services. For initializing the model, we need to generate PALM API.

Activity 1: Generate PALM API

Click on the link (<https://developers.generativeai.google/>). Then click on “Get API key in Google AI Studio”.

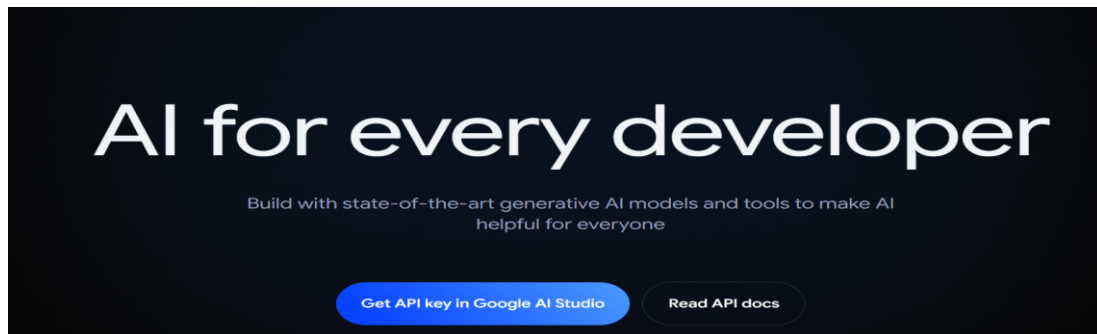


Fig 4.1.4 Google Generative AI page

Click on “Get API key” from the right navigation menu. Now click on “Create API key”. (Refer the below images) Copy the API key.

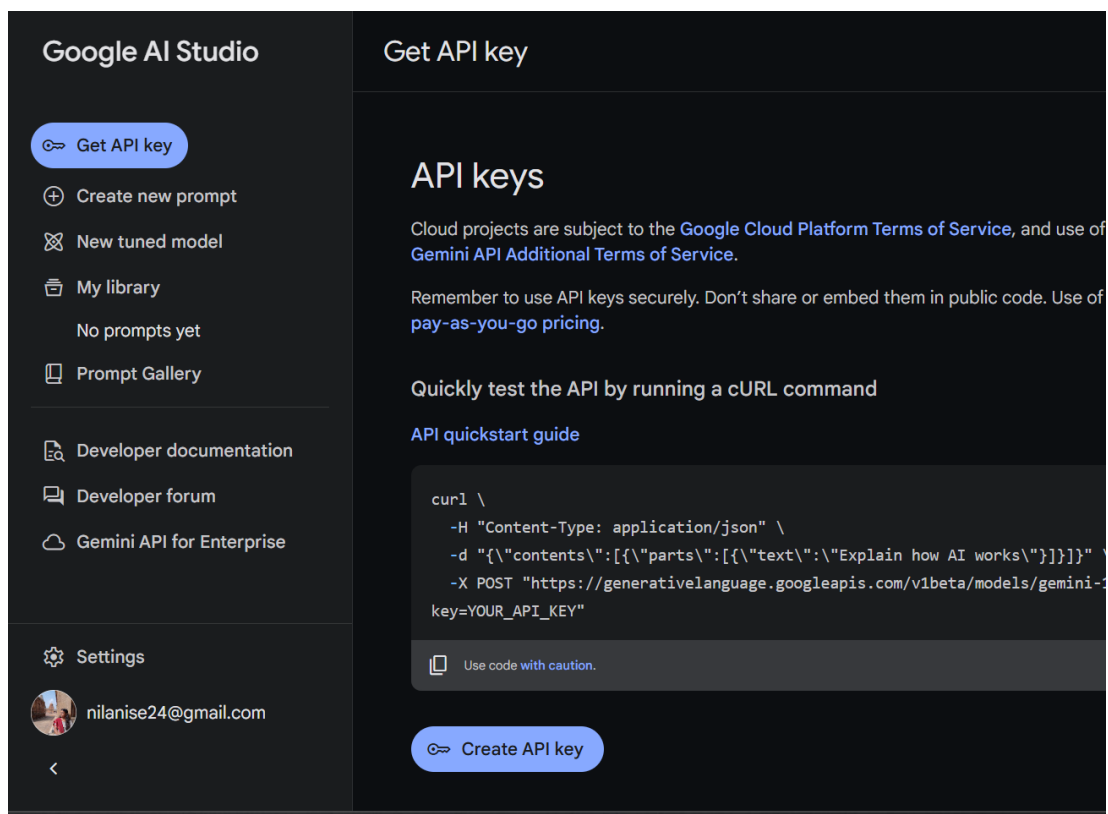


Fig 4.1.5 Google AI Studio Page

Next, click on 'Create API Key' and choose the generative language client as the project. Then, select 'Create API key in existing project'. Copy the newly generated API key as it is required for loading the pre-trained model.

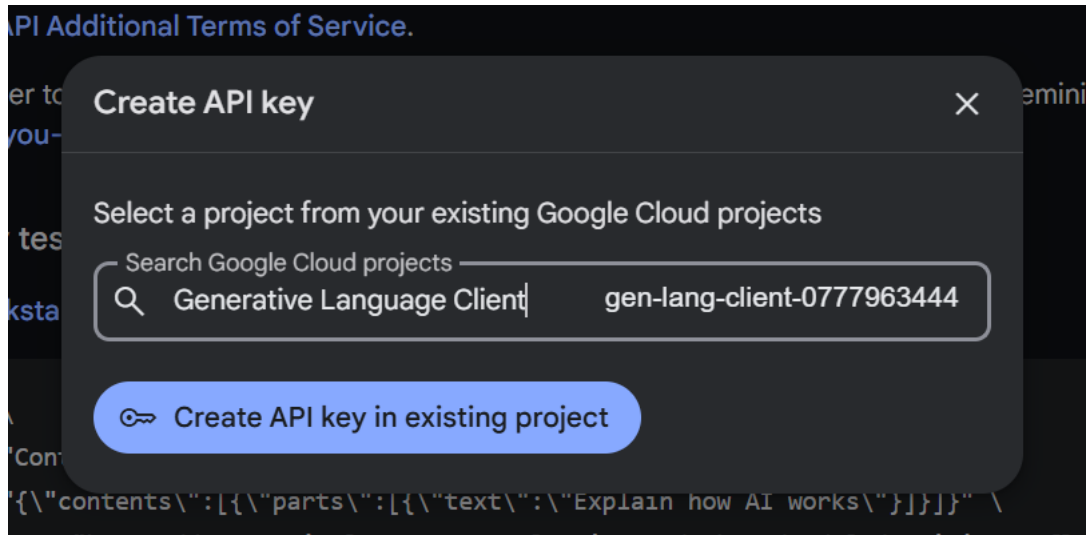


Fig 4.1.6 Selecting Generative Language Client

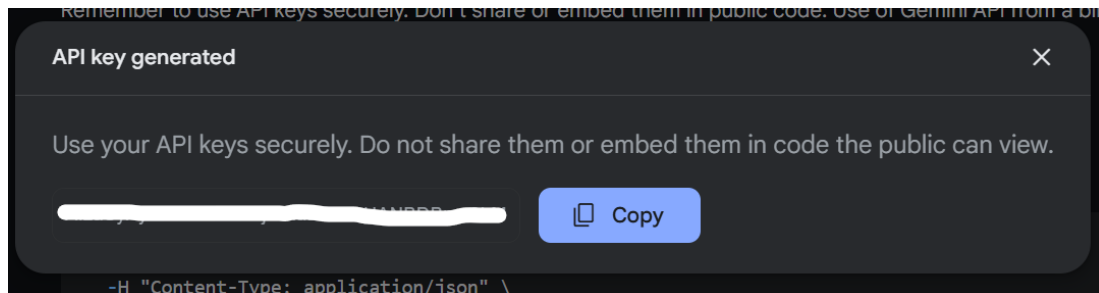


Fig 4.1.7 API Key Generated

Activity 2: Initialize the pre-trained model

Import Necessary Files

```
import streamlit as st
import google.generativeai as palm
import base64
```

Fig 4.1.8 Necessary Files

Streamlit, a popular Python library, is imported as `st`, enabling the creation of user interfaces directly within the Python script. Importing the `PaLM` module, imports the `PaLM` module from the `google.generativeai` package.

Configuration of the PALM API with the API key and initialize translator

```
#Configure the API with your API key
palm.configure(api_key="sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx")
```

Fig 4.1.9 Configure PaLM API key

Configuring the API key: The configure function is used to set up or configure the Google API with an API key. The Translator class facilitates language translation capabilities within the application.

Define the model to be used

```
#Define the model to use
model_name = "models/text-bison-001"
```

Fig 4.1.10 Model Name

the string "models/chat-bison-001", which identifies a specific model provided by Google's PaLM (Pathways Language Model) API. This variable is used in subsequent API calls to specify which model to use for generating responses based on user prompts.

To change the background and add pictures

```
page_bg_img = f"""
<style>
[data-testid="stAppViewContainer"] > .main {{
background-image: url("https://images.unsplash.com/photo-1501426026826-31c667bdf23d");
background-size: 180%;
background-position: top left;
background-repeat: no-repeat;
background-attachment: local;
}}

```

Fig 4.1.11 Add Background

Milestone 3: Interfacing with Pre-trained Model

In this milestone, we will build a prompt template to generate code based on user description and language.

Activity 1: Questions Generator

```
if st.button("Generate Itinerary"):
    #Placeholder for the generated itinerary
    itinerary = ""
    food_places = ""
    # Generate Itinerary using the selected model
    try:
        with st.spinner("Generating Itinerary..."):
            #Generate text using the model
            prompt = f"Generate an itinerary for a {num_days}-day trip to {destination} for the purpose of {purpose}."
            #Include details about nearby food places in {destination}."
            response = palm.generate_text(model=model_name, prompt=prompt)
            itinerary = response.result #Adjust this based on the actual response structure

    except Exception as e:
        #Display detailed error message
        st.error(f"Error generating itinerary: {e}")
        st.exception(e)
        st.warning("Please check your inputs and try again.")

    # Display the generated itinerary and food places

    if itinerary:
        st.success("Itinerary generated successfully!")
        st.text_area("Generated Itinerary:", value=itinerary,height=400)
    else:
        st.warning("No itinerary generated. Please try again with different inputs.")
```

Fig 4.1.12 Itinerary Generator

When the button is pressed, it initializes placeholders for the itinerary and nearby food places. It then constructs a prompt incorporating the user-provided destination and number of days. The prompt is sent to the AI model (specified by model_name) to generate the itinerary. During this process, a spinner is displayed to indicate that the system is working. If the generation is successful, the resulting itinerary is displayed in a text area. If an error occurs, detailed error messages are shown, and users are prompted to check their inputs and try again. This approach ensures a user-friendly experience, guiding users through generating and viewing their custom travel plans.

Milestone 4: Model Deployment

In this milestone, we are deploying the created model using streamlit. Model deployment using Streamlit involves creating a user-friendly web interface for deploying machine learning models, enabling users to interact with the model through a browser. Streamlit provides easy-to-use tools for developing and deploying data-driven applications, allowing for seamless integration of machine learning models into web-based applications.

Activity 1: Deploy the application using Streamlit

Give the project title , Description and Describe the scenarios

```
# Streamlit App Title
st.title("AI Travel Planner")

#

st.header("TripTrek: Intelligent Travel Planning with AI")
# Introduction text
st.markdown("""<div style="text-align: justify;"> TripTrek is an AI-powered travel planning platform designed to revolutionize the way people plan their trips. It leverages advanced AI algorithms to offer personalized itineraries and recommendations based on user preferences and input.

st.subheader("""Scenario 1: Family Vacation Coordination""")
st.markdown("""<div style="text-align: justify;"> TripTrek helps families plan their vacations by taking user inputs such as destination, travel dates, and budget, and providing tailored suggestions for activities, accommodations, and transportation.

st.subheader("""Scenario 2: Business Travel Planning for Professionals""")
st.markdown("""<div style="text-align: justify;"> TripTrek streamlines business travel for professionals by taking user inputs like destination, travel dates, and company policy, and offering optimized itineraries, hotel recommendations, and flight options.

st.subheader("""Scenario 3: Educational Trip for Students""")
st.markdown("""<div style="text-align: justify;"> TripTrek assists in planning educational trips for students by taking inputs like destination, travel dates, and budget, and providing suggestions for educational institutions, travel arrangements, and local experiences.
```

Fig 4.1.13 Web Application Content

The provided code introduces TripTrek, an AI-driven travel planning platform aimed at revolutionizing trip organization. It leverages advanced AI algorithms to offer personalized itineraries. Three scenarios illustrate TripTrek's capabilities: Family Vacation Coordination, Business Travel Planning for Professionals, and Educational Trips for Students, showcasing its ability to cater to diverse travel needs with tailored itineraries and recommendations.

Take user inputs

```
# User Input for Travel Destination
destination = st.text_input("Enter your travel destination: ", "Enter destination")

# User Input for Number of Days
num_days = st.number_input("Enter the number of days for your trip: ", min_value= 1, max_value=30, step=1)

# User Selection for purpose of trip
purpose = st.selectbox("Enter the purpose of travel", ('Select an option', 'Tourism', 'Business', 'Education'))
```

Fig 4.1.14 User Inputs

In this section of the Streamlit application, users are prompted to input their travel destination and the number of days for their trip. The `st.text_input` function is used to capture the destination as a text input from the user, allowing them to specify the location they plan to visit. The `st.number_input` function is then employed to obtain the number of days for the trip, with a minimum value of 1 day and a maximum of 30 days, allowing users to select the duration of their stay using a convenient stepper interface. These inputs are essential for generating a personalized travel itinerary, as they provide the necessary parameters for the AI model to tailor recommendations for activities, accommodations, dining options, and other travel-related details specific to the user's chosen destination and trip length.

Run The Web Application

```
PS C:\Users\senil\Hello-world> & C:/Users/senil/anaconda3/envs/myenv/python.exe c:/Users/senil/Hello-world/TRAVELPLANNER/app.py
2024-11-13 19:30:55.436
Warning: to view this Streamlit app on a browser, run it with the following
command:

streamlit run c:/Users/senil/Hello-world/TRAVELPLANNER/app.py [ARGUMENTS]
```

Fig 4.1.15 Run Web Application

Open the anaconda prompt from the start menu Navigate to the folder where your Python script is. Now type “`streamlit run app.py`” command Navigate to the localhost where you can view your web page

```
PS C:\Users\senil\Hello-world> streamlit run c:/Users/senil/Hello-world/TRAVELPLANNER/app.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.16.153.116:8501
```

Fig 4.1.16 Run in Streamlit

Now, the application will open in the web browser.



Fig 4.1.17 Web Application in Browser

5 Test Results:

5.1 Test cases:

TEST CASE ID	TEST CASE DESCRIPTION	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-01	Verify destination options display in dropdown	All available destinations are shown in dropdown	All destinations displayed correctly	Pass
TC-02	Select a destination successfully	Selected destination is saved and displayed	Destination saved and displayed in summary	Pass
TC-03	Select an invalid destination	System displays error message	Error message	Pass

			displayed for invalid input	
TC-04	Verify selected destination displays correctly in summary	Selected destination appears in trip summary	Destination shown correctly in summary	Pass
TC-05	Select start and end dates	Both dates are selectable	Dates selected successfully	Pass
TC-06	Calculate number of days between start and end dates	Number of days is correctly calculated	Calculation off by one day	Fail
TC-07	End date is before start date	Error message is displayed	Error message displayed for invalid date range	Pass
TC-08	Display travel purpose options (leisure, business, adventure)	All purpose options are shown	All options displayed correctly	Pass

TC-09	Select purpose of travel successfully	Selected purpose is saved and displayed	Purpose saved and shown in summary	Pass
TC-10	No travel purpose selected	Error message is shown	Error message displayed when purpose not selected	Pass
TC-11	Confirm trip details are saved successfully	Trip is confirmed and saved to system	Trip confirmation fails with a server error	Fail
TC-12	Attempt generate with incomplete data	Error message is shown, generation blocked	Error message displayed; generation blocked	Pass

6 Results & Discussion:

The Results of the TripTrek travel planner implemented in the above section includes:

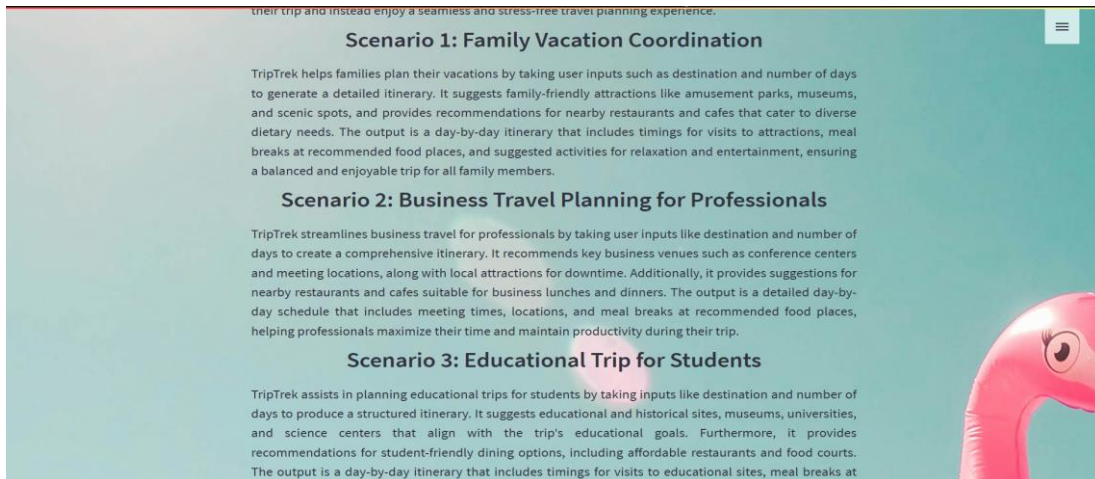


Fig 6.1 TripTrek Web Application

The screenshot shows the user input interface of the TripTrek web application. It features three input fields: "Enter your travel destination:" with a text input field containing "Enter destination"; "Enter the number of days for your trip:" with a numeric input field containing "1" and a "+" button; and "Enter the purpose of travel" with a dropdown menu showing "Select an option". Below these fields is a "Generate Itinerary" button. A pink flamingo illustration is visible on the right side of the page.

Fig 6.2 User input interface

Output For Tourism travel in Chennai for two days,

The screenshot shows a web interface for generating a travel itinerary. At the top, there are three input fields: 'Enter your travel destination:' with 'Chennai' entered, 'Enter the number of days for your trip:' with '2' entered, and 'Enter the purpose of travel:' with 'Tourism' selected from a dropdown menu. Below these fields is a red button labeled 'Generate Itinerary'. A green banner below the button says 'Itinerary generated successfully!'. Underneath, the 'Generated Itinerary:' section is displayed. It starts with '**Day 1:**' and is divided into '**Morning:**' and '**Afternoon:**' sections. The morning section includes: 'Arrive at Chennai International Airport (MAA)', 'Take a taxi to your hotel (The Park Chennai, The Leela Palace Chennai, or The Taj Coromandel Chennai are all good options)', 'Check in to your hotel and leave your luggage', 'Head to the Marina Beach for a walk and to watch the sunset', and 'Have dinner at a nearby restaurant (try the seafood!)'. The afternoon section includes: 'Visit the Kapaleeshwarar Temple, a Hindu temple dedicated to Shiva', 'Take a walk through the Mylapore neighborhood, which is home to many temples and churches', 'Have lunch at a local restaurant (try the idli and dosa!)', and 'Visit the Chennai Museum, which houses a collection of artifacts from the city's history'.

Enter your travel destination:
Chennai

Enter the number of days for your trip:
2

Enter the purpose of travel
Tourism

Generate Itinerary

Itinerary generated successfully!

Generated Itinerary:

****Day 1:****

****Morning:****

- * Arrive at Chennai International Airport (MAA)
- * Take a taxi to your hotel (The Park Chennai, The Leela Palace Chennai, or The Taj Coromandel Chennai are all good options)
- * Check in to your hotel and leave your luggage
- * Head to the Marina Beach for a walk and to watch the sunset
- * Have dinner at a nearby restaurant (try the seafood!)

****Afternoon:****

- * Visit the Kapaleeshwarar Temple, a Hindu temple dedicated to Shiva
- * Take a walk through the Mylapore neighborhood, which is home to many temples and churches
- * Have lunch at a local restaurant (try the idli and dosa!)
- * Visit the Chennai Museum, which houses a collection of artifacts from the city's history

Fig 6.3 Itinerary for Tourism

Output for Business travel in Chennai for two days,

The screenshot shows the same web interface as Fig 6.3, but with 'Business' selected in the 'Enter the purpose of travel:' dropdown menu. The 'Generate Itinerary' button is still red. The green banner says 'Itinerary generated successfully!'. The 'Generated Itinerary:' section starts with 'Day 1:' and is divided into '**Morning:**' and '**Evening:**' sections. The morning section includes: 'Arrive at Chennai International Airport (MAA)', 'Take a taxi to your hotel, The Park Chennai', 'Check in to your hotel and leave your luggage', and 'Head to the nearby restaurant, Saravana Bhavan, for breakfast'. The evening section includes: 'Attend a business meeting at the Taj Coromandel Hotel' and 'Have dinner at the hotel restaurant'. The afternoon section is empty.

Enter your travel destination:
Chennai

Enter the number of days for your trip:
2

Enter the purpose of travel
Business

Generate Itinerary

Itinerary generated successfully!

Generated Itinerary:

Day 1:

****Morning:****

- * Arrive at Chennai International Airport (MAA)
- * Take a taxi to your hotel, The Park Chennai
- * Check in to your hotel and leave your luggage
- * Head to the nearby restaurant, Saravana Bhavan, for breakfast

****Evening:****

- * Attend a business meeting at the Taj Coromandel Hotel
- * Have dinner at the hotel restaurant

Fig 6.4 Itinerary for Business

Output for Education purpose travel in Hyderabad for two days,

The screenshot displays the Triptrek travel planner interface. It features a form with three input fields: 'Enter your travel destination:' with 'Hyderabad' entered, 'Enter the number of days for your trip:' with '2' entered, and 'Enter the purpose of travel' with 'Education' selected from a dropdown menu. A red 'Generate Itinerary' button is positioned below the form. A green banner with the text 'Itinerary generated successfully!' is visible. Below the banner, the 'Generated Itinerary:' section shows the following details:

- **Day 1:****
- * **Morning:****
- * Arrive at Rajiv Gandhi International Airport (HYD)
- * Take a taxi to your hotel (Hotel Novotel Hyderabad Airport, 1-200, Gachibowli, Hyderabad, Telangana 500032, India)

Fig 6.5 Itinerary for Education

Triptrek, the AI-powered travel planner, has the potential to revolutionize the travel industry. It can be used by individuals to create personalized itineraries, book flights and hotels, and receive real-time recommendations. Travel agencies and tour operators can leverage Triptrek to enhance services, streamline operations, and gain a competitive edge. Destination marketing organizations can use it to promote destinations, manage tourist flows, and make data-driven decisions. Businesses and corporations can benefit from Triptrek for corporate travel management, event planning, and incentive travel. By harnessing the power of AI, Triptrek can provide efficient, personalized, and data-driven travel solutions for a wide range of users and industries.

7 Conclusion & Future Work:

7.1 Conclusion:

In conclusion Triptrek, an AI-powered travel planner, has the potential to revolutionize the way we travel. By leveraging advanced technologies like machine learning, natural language processing, and augmented reality, Triptrek offers a personalized, efficient, and enjoyable travel experience. From generating tailored itineraries to providing real-time recommendations and support, Triptrek empowers travelers to make informed decisions and explore the world with ease. As technology continues to evolve, Triptrek can further enhance its capabilities, offering even more innovative features and services. By embracing AI and focusing on user experience, Triptrek aims to become the ultimate travel companion for the modern traveler.

7.2 Future Work:

Triptrek can be enhanced through deep learning to adapt itineraries in real-time based on user preferences and evolving conditions. By leveraging behavioral insights, the platform can offer increasingly personalized and relevant travel experiences. AR and VR integrations can provide virtual previews of destinations and accommodations, while predictive analytics can forecast prices and crowd levels to optimize planning. The platform can promote sustainable travel by highlighting green options and offering carbon footprint tracking. Real-time data integration ensures up-to-date information on weather, traffic, and travel alerts. Multilingual support and conversational AI capabilities make Triptrek accessible to a global audience. Mobile-first, offline access, and in-trip support cater to the needs of modern travelers. Voice and gesture interfaces enhance user interaction, while user-generated content fosters a collaborative travel community. Advanced data privacy and security measures protect user information, ensuring a safe and secure travel planning experience.

8 References:

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